exceed the export bill of electronics in India. At some point we might land up with a $100 billion trade deficit in terms of electronics.

Being dependent on outside for something that is as strategic as semiconductors is not a good idea. An economy that is as aspirational and as big as India's should have access to a spectrum of leading technologies. We should focus on semiconductors not only because we want to be self-reliant in terms of semiconductors, but also because it could give us access to a spectrum of technologies.

With political stability as one of the biggest advantages of India, do you think the country can become the hub of semiconductor manufacture?

Asia's demand is already being met by the suppliers to a large extent; the likes of Taiwan, Japan, Korea, and China are meeting semiconductor demands quite successfully. We might not be able to get very far by trying to replace the existing suppliers of semiconductors. What we could do probably better in India is to generate a whole new range of applications for semiconductors. India has a unique set of problems where millions are transitioning from a farm-based economy to an urban-based economy. The problems during this transition are going to be both a challenge and an opportunity. Technology will play a big role in charting out solutions for these problems.

The best part is that, once these solutions are available, the same will find applications in a lot of other parts of the world. I see the continent of Africa and Latin America as two of the biggest examples. Some examples of the solutions could be concerning agriculture, weather forecast, and communication. Instead of replacing current leaders in terms of semiconductors, we should try and find solutions to newer problems, and then proliferate the same to the rest of the world.

What kind of semiconductor fab do you think India should look at under the present circumstances?

That is a difficult question to answer. Instead of looking at what technology a country should start with in terms of a semiconductor fab, it should look at what applications it wants to focus on. In my 25-year journey with Lam, I have gone from 250Nm technology to 5Nm technology.

What we at Lam have recently discovered is the repurposing of older technologies for newer applications. For example, if you are building an IoT device that generally does not require cutting-edge technology, or a water sensor for assessing the quality of water in a lake or a pond, it would not require 5Nm technology. These can be developed using 90Nm or 65Nm technology.

It is the application that should drive selection of the technology, and hence the kind of investment required. The expense in setting up a 5Nm fab is humongous. Instead of debating whether to go for 12Nm or 5Nm technology, we should look at the applications we want to build. EFY